



COR 601: Principle of Circular Economy

Academic Year 2025/26 Term 2

Title:

Secondary Markets: A Sustainable Solution or a Driver of Overconsumption?

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Introduction

Sustainability is practiced in every industry due to increasing concerns on environmental harm caused by the industries that pollute land, water and air. It has become a norm for every industry to adopt circular strategies such as reuse and recycling to lessen environmental, social and economic harm. In the context of apparel industry, polyester, polyamide and acrylics are the significant portions of fabric used in production, that pose sustainability challenges (Anceschi, et al., 2026).

Secondary markets are being developed and expanded massively by various stakeholders including manufacturers, to lengthen the lifespan of every product that is available in the market. It is expected to grow double in market size by 2027, from \$177 billion in 2022 (Mizrachi & Sharon, 2025). Such markets have adopted resale and rental models to effectively lessen the generation of waste. Through the literature review conducted by Anceschi, et al. (2026), it was found that reuse and mechanical recycling are considered the most favourable methods of extending the life of products, compared to incineration, thermal recycling and chemical recycling.

This paper argues that the role of secondary markets has rather led to over-consumption of apparel products.

Consumer behaviour

Consumer behaviour is analysed through the lens of rebound effects and moral licensing. Rebound effects can be understood through CER (Circular Economy Rebound), where anticipated environmental gains are diminishing or even reversed due to systemic, market, or behavioural responses (Lowe, et al., 2025). This challenge is owing to growth-oriented economy that prioritises resource efficiency without reducing overall consumption. In recent literature, CER is associated with lower impact of per unit production, but has amplified the overall production and consumption rather than decreasing it.

Rebound effects are usually difficult to be mitigated because there will be expanded consumption when prices become lower owing to market efficiency. Rebound effects occur not only due to economic growth but also capital accumulation. Such effects cannot be reduced merely through taxes and subsidies but require radical forms of intervention i.e., restrictions on production, resource extraction, or consumption. Emphasis should not be given only to carbon metrics (negative outcomes) but also on how policy makers, businesses and the public can together consider CER while staying inter-connected.

The firms have fewer incentives to assess CER (i.e., to quantify and disclose them) and it is beyond the firm's boundary, so it does not fall under the realm of corporate decision making. Firms view the rebound effect as a positive outcome because the resulting spike in consumption directly boosts their profits. This financial incentive creates a conflict of interest, as CER simultaneously hampers genuine sustainability efforts.

Consumer trends between 2020 and 2024 reveal a significant shift toward less sustainable habits, despite the growth of secondary markets (Mizrachi & Sharon, 2025). There has been a marked increase in online shopping and a surge in purchases from ultra-fast fashion platforms. Consumers are also buying new clothing at historically low prices more frequently than in the past. Conversely, clothes-swapping and reselling have declined. These shifts support the argument that the growth of secondary markets may be encouraging, rather than reducing, total global consumption.

There is moral licensing attached to the purchase of used clothes or donation of clothes. This drives people to buy a lot more new or used clothes at both primary and secondary cloth markets, as they are guilt free and have the notion that there is lesser damage to the environment through their actions. Brooks (2025) emphasises that to be sustainable, consumers must re-use rather than recycle. If they re-use, there would be reduction in the purchase of new clothes, which in turn would lead to a reduction in the volume of such clothes to be disposed or exported to lower-income countries.

This is further reflected in actual consumer data. Koch et al. (2025) found that when fewer new clothes are bought and durable garments are chosen, it had the highest environmental reduction potential, while the second-hand purchasing and renting clothing ranked the lowest. Their estimates indicate that when consumers in Netherlands make sustainable choices, it reduces 11-19 % of greenhouse gas emissions and nearly 3-29 % usage of land for clothing purposes. Dutch consumers who buy clothes on rental comprise of only 1% and about 4% will be willing to adopt sustainable behaviour under ideal conditions. In the context of buying second hand clothes, behavioural plasticity is about 23% where only 3% currently buy such clothes. These numbers are very low, which questions how much secondary markets can contribute to reducing environmental harm in practice.

Global Supply Chain

Global North consumes clothing excessively and discards it, which then is exported to the Global South. The second-hand clothing is not just imported to fulfil the demands of local

consumers, but it is rather dumped. Supply of such clothing exceeds the demand of local consumers; and is a serious concern. On top of that, Global South is the most resource-intensive manufacturing hub, which affects their environment worse; and their limited knowledge on sustainable practices, makes it even more worse.

Through the research conducted by Bonelli et al. (2024), it is found that responsibilities and burdens towards climate change mitigation effort are unequally distributed between the Global North and Global South due to presence of inequalities that is ingrained in economic and political structures. Such inequalities make the production and consumption patterns differ between these two broader regions.

Fast fashion as a business model is unsustainable from both the demand and supply side (Bonelli, et al., 2024). So, what has driven such massive amounts of clothing to be dumped in the Global South? There is overproduction largely due to consumer behavioural shifts from the 1990s. The companies started to sell products at a lower cost with an intent to use and discard as a fast fashion trend. Though the consumers are aware of such sustainable practices and are concerned over environmental harm, they would still tend to buy more clothes also due to the presence of secondary markets. The gap between consumer awareness of environmental harm and actual sustainable purchasing behaviour is widely noted in the literature (Bonelli et al., 2024).

UN Comtrade data was used in the study, where Brooks (2025) finds that the used clothes exported in 2024 equalled 4,848,586,806 kg. The author mentions that not all clothes being imported into the Global South are re-used by poor people. But rather, a lot of such clothes are highly toxic waste and are being disposed through various means. It is dumped on the land, thrown into water streams, incinerated, etc. There is no proper infrastructure being invested in, to recycle the used clothes in such regions. Brooks (2025) demonstrated that textile waste is not reduced but rather it is pushed downstream to poorer countries. Exporting of used clothes does not necessarily lead to lower carbon emissions either because transportation of these used clothes generates emissions as well; what is rather needed is a reduction in new clothing consumption and a shift toward slow fashion. Lastly, the trade does not create jobs that are well-paid because such imports consist of clothes that have lesser market value. Such an international second-hand trade is worsening Global South not just environmentally, but also culturally and economically.

The anti-poverty organizations like Oxfam which aim to eradicate poverty have worsened the inequality by actively participating in the second-hand international trade. Secondary markets are structurally flawed as well.

Life Cycle Assessment

LCA results cannot be generalised when researchers and practitioners rely on under-representative data or analyse too few products (Makov & Vivanco, 2025). It is because LCA differs ‘within and between’ products as it is based on the product model, weight, material composition, assessment method used, etc. However, when analysis is carried out on the mix of products where their characteristics and market share is considered within the segment, it is a more reliable estimate.

Aneschi, et al. (2026) critically review the methods: Life Cycle Assessment (LCA), Life Cycle Costing (LCC), and Social Life Cycle Assessment (S-LCA). LCA is the most widely adopted tool compared to LCC and S-LCA. One of the key challenges in utilizing LCA in the textile industries are inconsistent data collection, unclear system boundaries, different business functions, varying stages of production, etc. Currently, LCA results are unreliable and cannot be compared across the studies in textile industry. The authors also note that some studies may not consider end-of-life cycle of the products when analysing its impact using LCA. This is particularly problematic in the context of secondary markets, where environmental claims are frequently made without reliable evidence.

The three methods LCA, LCC and S-LCA are used in isolation and is quite challenging to integrate to understand the impact of textile industry across environmental, economic, and social dimensions. In second-hand markets, environmental benefits are often reported alongside hidden social and economic costs that go unexamined. Few studies incorporate S-LCA to examine social dimensions like workers’ rights, labour conditions, community well-being, etc; but consumers are least studied on. Standardization and transparency of these methods along with other approaches is emphasized by the authors to tackle the challenges of sustainability. It improves the scientific robustness of sustainability assessments. It is crucial that these assessments cover the full life cycle from raw material extraction to end-of-life disposal.

These limitations are evident when LCA is applied to secondary markets specifically. Gray et al. (2025) used LCA to compare five clothing circular business models (online resale, commercial resale, charity resale, short-term rental and longer-term leasing) to determine

whether these models increase or decrease the environmental impact compared to the purchase of newly manufactured clothes. It was found that it led to 14-26% reduction in greenhouse gas emissions, where online resale ranked the highest and short-term rental the least. It was the displacement rate that majorly drove these results. They had not calculated the rebound effect and is not reflected in such positive results, which raises questions on whether these results are reliable and whether it reflects the true environmental impact of the garments in the apparel industry. Conducting more studies in the future is recommended by the authors to understand whether these results are truly the case. It must be noted that increasing adoption of the resale models might reverse the reduced environmental impact compared to new clothing business models, if these different models are not operationalised strategically. Ultimately, buying second-hand must replace new clothing purchases, not add to them, to reduce environmental impact.

Conclusion

Secondary markets are promoted as a sustainable solution in the apparel industry, but this paper argues that they have rather led to overconsumption. Consumer behaviour driven by rebound effects and moral licensing has resulted in people buying more clothes, not less. The export of used clothing from the Global North to the Global South has not led to any kind of development of Global South, it has only shifted the environmental and economic burden onto poorer nations. The assessment tools used to measure the sustainability of these markets are inconsistent and unreliable, making it hard to trust the environmental claims made about circular fashion. More reliable assessment tools are needed that capture the true impact of apparel products across their entire life cycle.

Buying fewer clothes and choosing durable, recycled materials is far more effective than expanding secondary markets. For secondary markets to truly contribute to sustainability, they must replace new clothing purchases entirely rather than add to them. Circularity in the apparel industry in practice requires a fundamental shift in consumption behaviour, not simply donating or exporting excess clothing to poorer countries.

References

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Name of AI Tool	Claude AI
Input Prompt	can you fix grammatical errors , improve flow and clarity, can you help me reduce about 25 words
Date Generated	27/ 03/2026
Output Generated	Introduction Fix 1: <i>"It has become a norm for every industry to adopt circular strategies (reuse, recycle); and make" → "It has become a norm for every industry to adopt circular strategies such as reuse and recycling, and make"</i> Fix 2: <i>"polyester, polyamide and acrylics are the significant portions of fabric that are used for production, that pose" → "polyester, polyamide and acrylics are significant portions of fabric used in production, posing"</i> Consumer Behaviour Fix 3: <i>"Rebound effects is usually difficult" → "Rebound effects are usually difficult"</i> Fix 4: <i>"but requires radical forms" → "but require radical forms"</i> Fix 5: <i>"This is further reflected in actual consumer data. Koch et al."</i> → Keep as is 

Global Supply Chain

Fix 6 — Delete this sentence entirely: *"It is thought that such actions are sustainable but rather are not."*

Fix 7: *"which affects their environment worse; and they are not knowledgeable in terms of sustainable practices, which makes it even more worse"* → *"which worsens their environmental situation, and their limited knowledge of sustainable practices compounds this further"*

Fix 8: *"The second-hand clothing is not just imported to fulfil the demands of local consumers, but it is rather dumped."* → *"Secondhand clothing imported into the Global South does not simply fulfil local demand — it is excess supply that is effectively dumped."*

Fix 9: *"There is overproduction largely due to consumer behavioural shifts from the 1990's"* → *"There is overproduction largely due to consumer behavioural shifts from the 1990s"*

Fix 10: *"with an intend to use and discard"* → *"with an intent to use and discard"*

Fix 11: *"The author mentions that not all clothes being imported into the Global South is being re-used"* → *"Not all clothes imported into the Global South are reused"*

Fix 12 — Delete this sentence: *"The actions taken on the lines of sustainability by importing used clothing into Global South, have negatively impacted such regions."*

LCA Section

Fix 13: *"LCA results cannot be generalised when researchers and practitioners rely on under-representative data or analyse too few products"* → Keep as is 

Fix 14: *"One of the challenges in utilizing LCA is that in the textile industries, there is inconsistent data collection, unclear system boundaries, different business functions, stages of production, etc."* → *"One of the key challenges in utilising LCA in the textile industry*

is inconsistent data collection, unclear system boundaries, and varying stages of production considered across studies."

Fix 15: *"Few studies incorporate S-LCA to study about social dimensions" → "Few studies incorporate S-LCA to examine social dimensions"*

Fix 16: *"They had not calculated the rebound effect and is not included in such positive results" → "The rebound effect was not calculated and is therefore not reflected in these positive results"*

Fix 17: *"whether it reflects the true environmental impact" → "whether they reflect the true environmental impact"*

Conclusion

Fix 18: *"The export of used clothing from the Global North to the Global South has not led to any kind of development of Global South" → "The export of used clothing from the Global North to the Global South has not contributed to the development of poorer nations"*

Cut 1 — Introduction

Current: *"It has become a norm for every industry to adopt circular strategies (reuse, recycle); and make a sincere effort to lessen the negative impact environmentally, socially and economically."*

Change to: *"It has become a norm for every industry to adopt circular strategies such as reuse and recycling to lessen environmental, social and economic harm."*

Saves 8 words

Cut 2 — Global Supply Chain

Current: *"It is thought that such actions are sustainable but rather are not."*

Delete entirely

Saves 13 words

	Cut 3 — LCA Section Current: <i>"It is crucial that these assessments cover the full life cycle from raw material extraction to end-of-life disposal to gain a comprehensive understanding of product's impact throughout its entire life cycle."</i> Change to: <i>"It is crucial that these assessments cover the full life cycle from raw material extraction to end-of-life disposal."</i> Saves 10 words Total Saved Approximately 31 words — slightly more than needed so you can keep one if you prefer.
Impact on Submission	